Robot Procedure 2015_011

Sep 22nd

2015

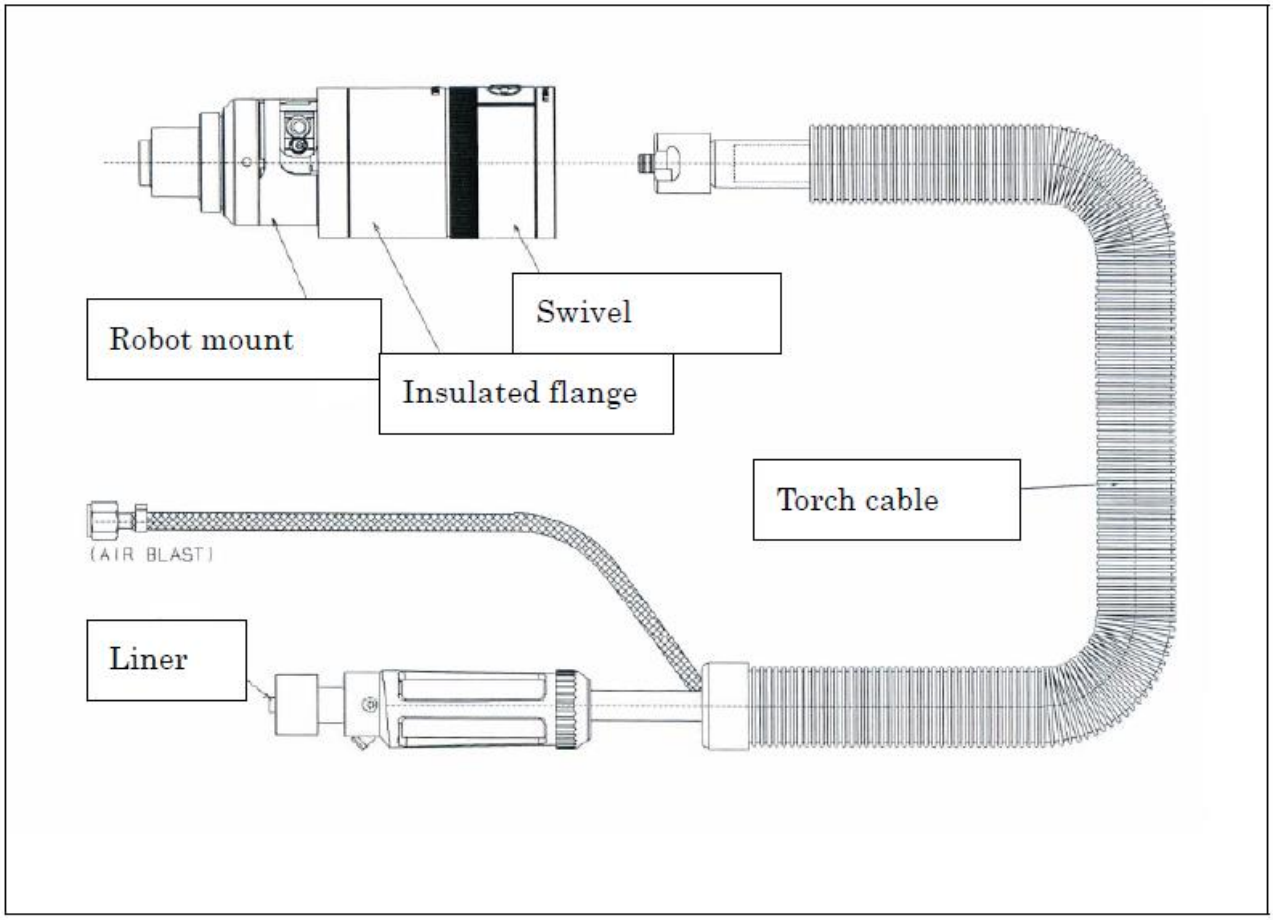
Procedure for Torch Cable, Liner, Neck & Tip Replacement

[Applicable for all Arc Weld Robots with <u>Binzel Touch</u> and Cable]

9.2 REPLACING TORCH CABLE, LINER, DRIVE ROLL, TORCH NECK

After replacing a part, be sure to check the corresponding item(s).
The table below indicates replacement parts and the corresponding check items. Make checks according to the table below.

Replacement part	Check item
Torch cable	Mating of the torch cable nut Liner replacement Wire feed capability
Liner Drive roll Wire guide	Wire feed capability
Torch neck	Mating of the torch cable nut Wire feed capability



9.2.1 Replacing torch cable

1 Move the robot posture to J4=0°, J5=-90°, J6=0° The J1-J3 axis is not cared about by arbitrary posture.
2 Remove the flexible conduit of wire feeder, cut the wire. Pull out the wire from the tip point after it supplies it until the wire cannot be sent. (It is not necessary to remove torch neck.)
3 Remove the conduit cover of J3 arm referring to Fig.9.2.1 (b).
4 Loosen the fixation two bolt of torch side referring to Fig.9.2.1 (c).
5 Remove torch cable from torch adapter.
6 Loosen the bolt with hexagon hole of wire feeder referring to Fig.9.2.1 (d).
7 Pull out the torch cable from the J3 arm part with Conduit.
8 Insert new liner to torch cable and tighten liner suppression nut.
9 Insert a new torch cable in the J3 arm part.
10 Match tip of metal of torch to ditch of key and insert it into torch adapter.
11 Confirm the bottom of tip of metal of torch cable has stuck to the face of torch adapter. When the torch cable doesn't enter easily, it becomes easy if a minus driver is inserted in the ditch, and the ditch is expanded. (See Fig.9.2.1 (f))
12 Shut up a conduit till it do not move towards the torch neck, and lock a conduit fixation cover
13 Loosen a fixed bolt of the wire feeder to the torch cable for the wire feeder to become parallel to wire feeder, tighten the bolt with hexagon hole of wire feeder referring to Fig.9.2.1 (d).
14 Attach fixation 2 bolts of torch side referring to Fig.9.2.1 (c).
Tighten black bolt with torque of 7.5Nm.
15 Firmly push Conduit into the interior before applying the Conduit fixation cover, and install the cover that fixes Conduit referring to Fig. 9.2.1 (b).
16 Lift the roll, insert wire and restore them.
17 Adjust the length of the liner referring to Subsection 9.2.2.

Fig.9.2.1 (a) Remove flexible conduit

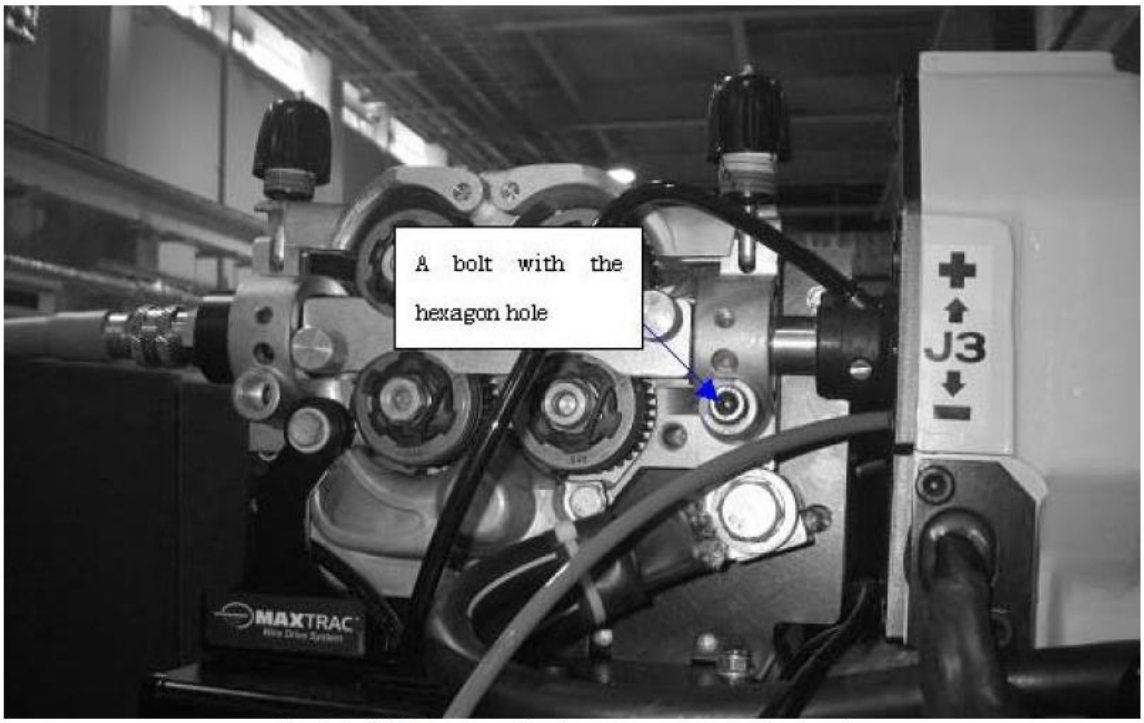


Fig.9.2.1 (d) Remove the thumbscrews of wire feeder

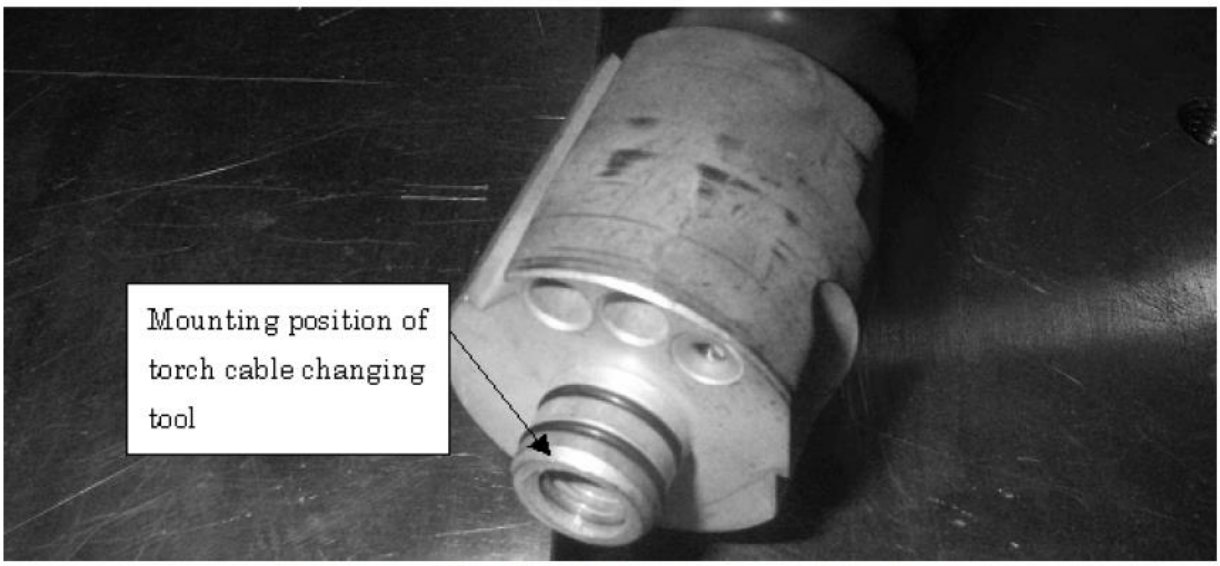


Fig.9.2.1 (e) Mounting position of torch cable changing tool

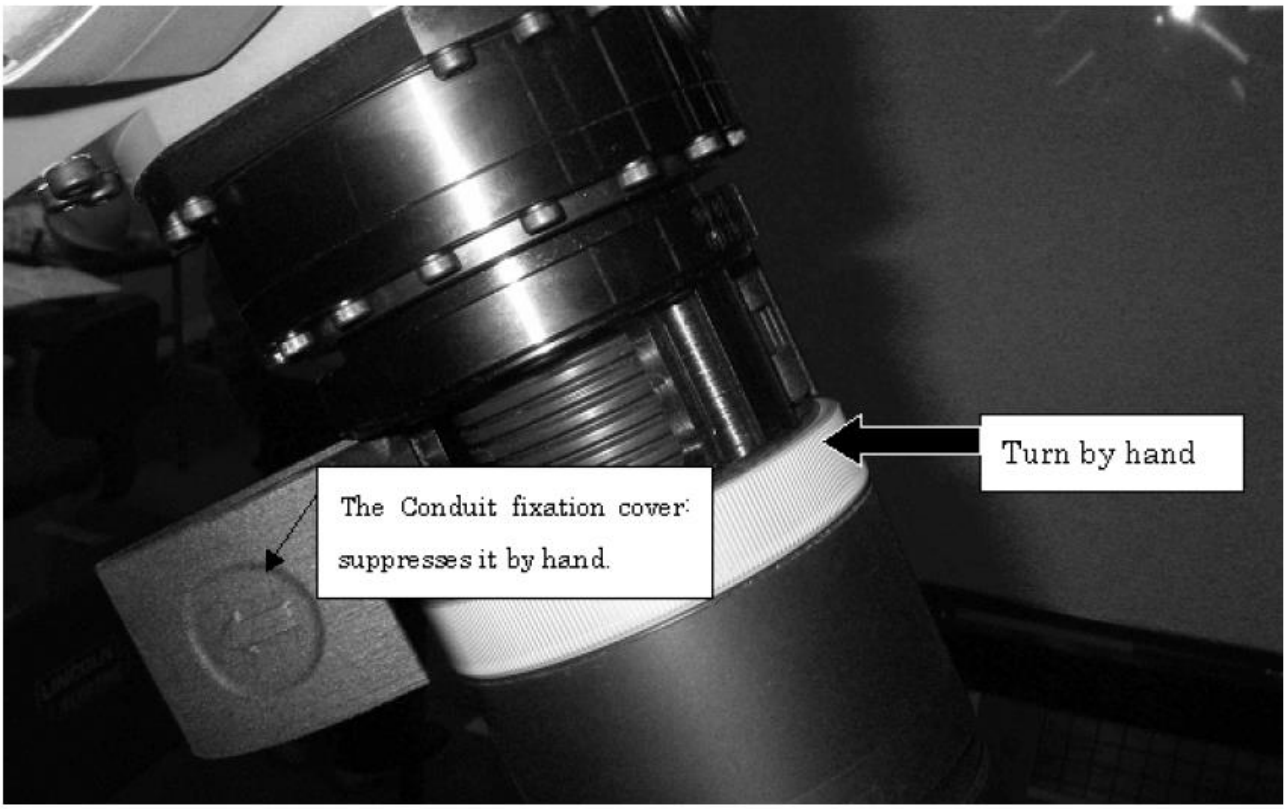


Fig.9.2.1 (b) Remove conduit

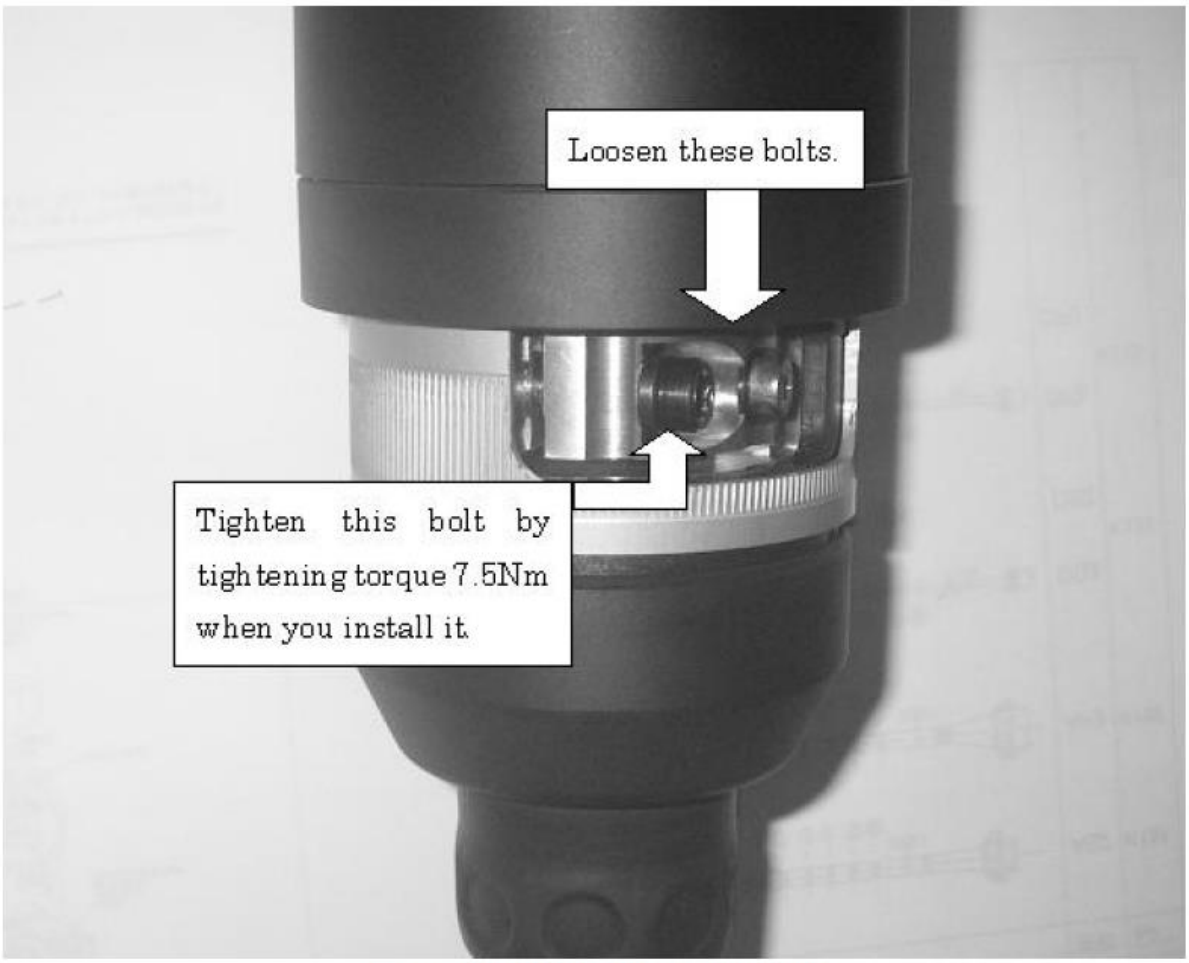


Fig.9.2.1 (c) Fixation bolt of torch side

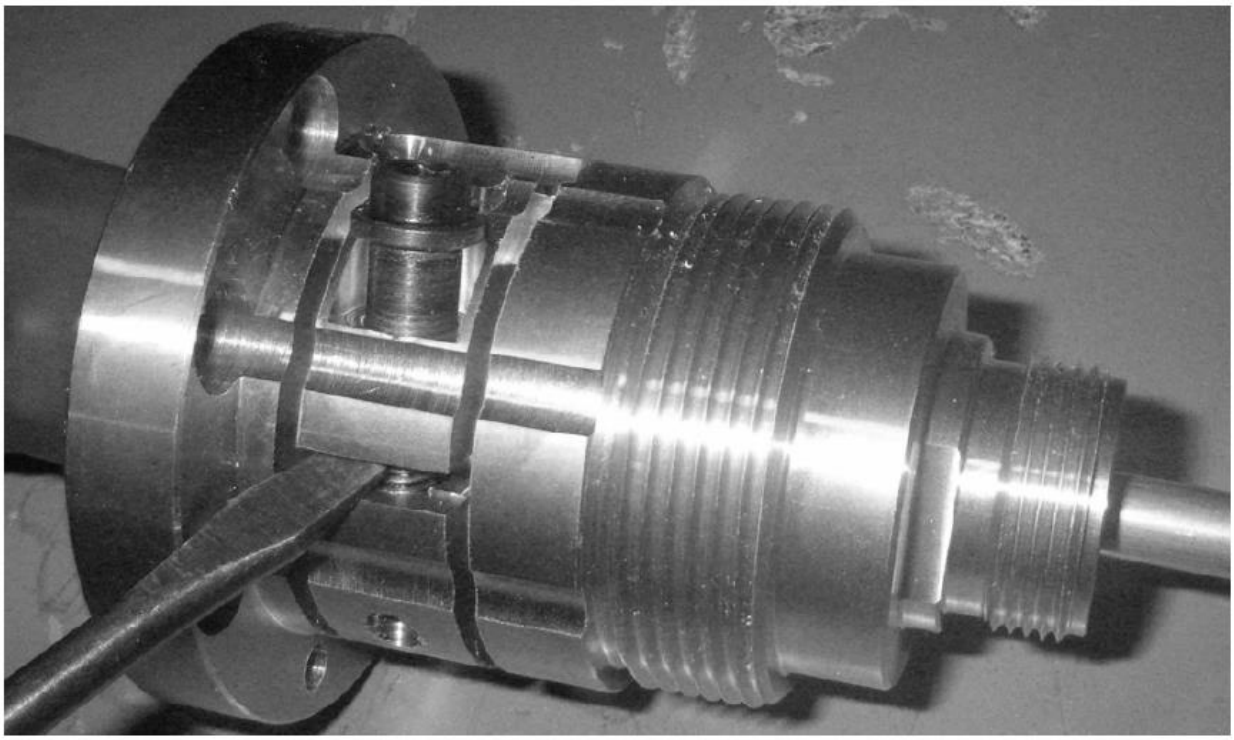


Fig.9.2.1 (f) Ditch of torch adapter

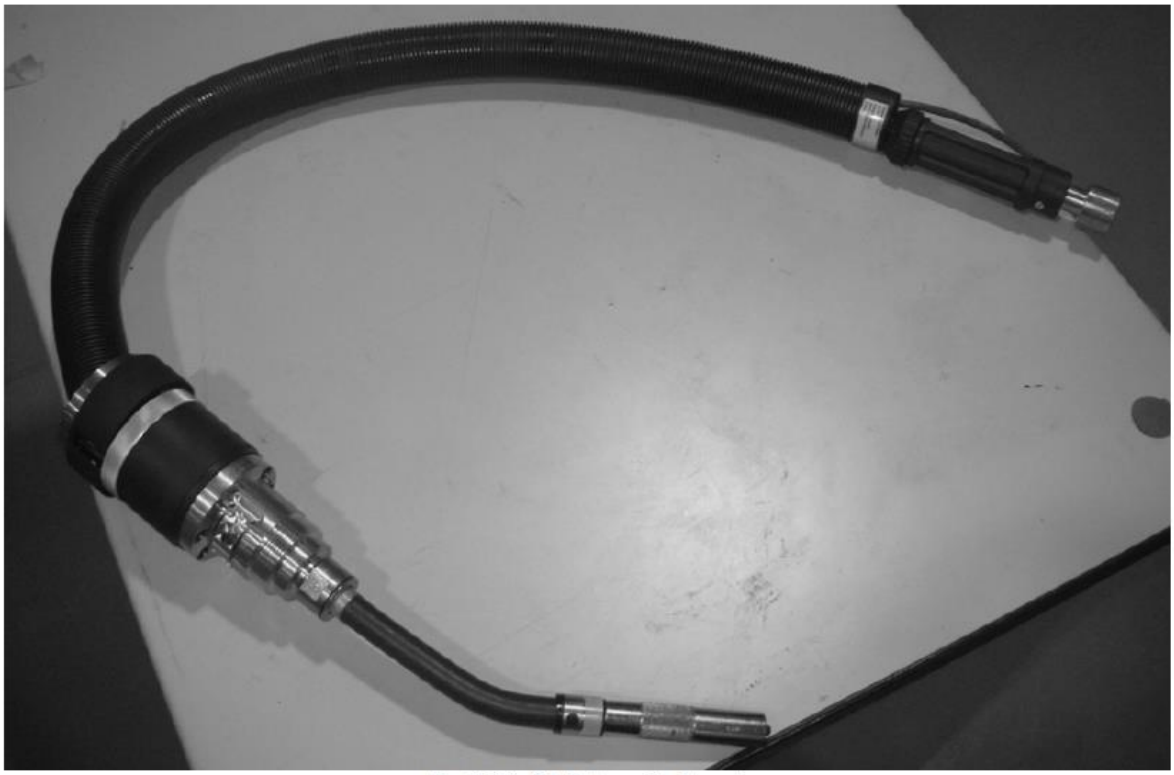


Fig.9.2.1 (g) External of torch

9.2.2 Replacing liner

1 Move the robot posture to J4=0°, J5=-90°, J6=0°. The J1-J3 axis is not cared about by arbitrary posture. And turn off the controller power and welding power supply.
2 Cut the wire around the roller of wire feeder, pull out the wire from the tip point.
3 Remove nozzle, gas diffuser and welding tip. At this time, pay attention not to drop and destroy gas diffuser from nozzle. (See Subsection 6.1.3 about each part.)
4 Loosen the bolt with the hexagon hole of wire feeder, and pull out torch cable from wire feeder.
5 Pull out the torch cable from the hole of J4-axis hollow parts of J3 arm to wrist side.
6 Loosen the liner suppression nut by the monkey spanner and remove it. (See.Fig.9.2.2 (a))
7 Pull out the liner from torch cable.
8 Insert liner to torch cable, and temporally tighten the liner suppression nut.
9 Cut the liner in the tip holder point.(See Fig.9.2.2 (b))
10 Pull out the liner from the torch cable again and cut about 8-10mm from the point further.
11 Peel off the liner hull (orange) from the point part to about 250mm. (See Fig.9.2.2 (c))
12 Remove burr, etc of cutting part. (Apply roundness.)
13 Insert liner to torch cable again, and tighten the liner suppression nut firmly.
14 Insert torch cable to J4-axis hollow part and insert torch into the interior while pushing bolt with the hexagon hole of wire feeder and tighten it ,and attach torch cable to wire feeder.
15 Turn on the controller power then feed the wire to check that the wire is fed through the torch tip. Next, operate the robot slowly to check that no problem is found.

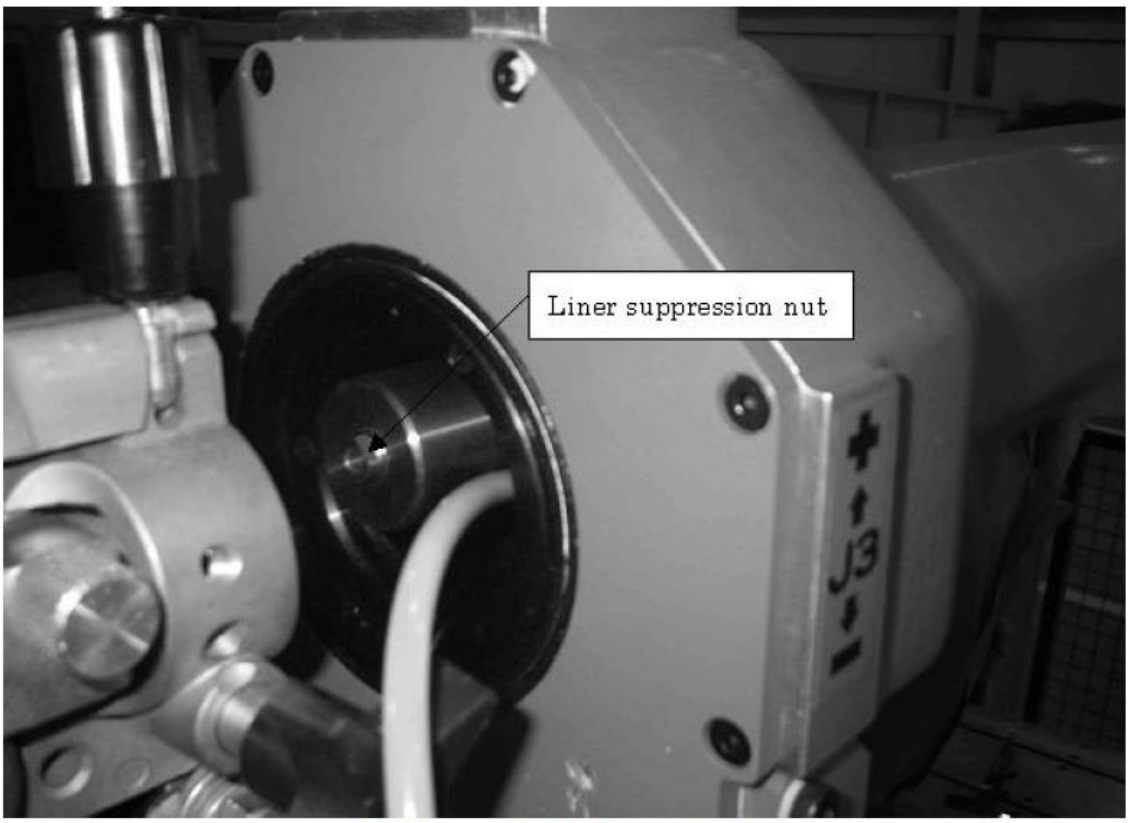


Fig.9.2.2 (a) Liner suppression nut

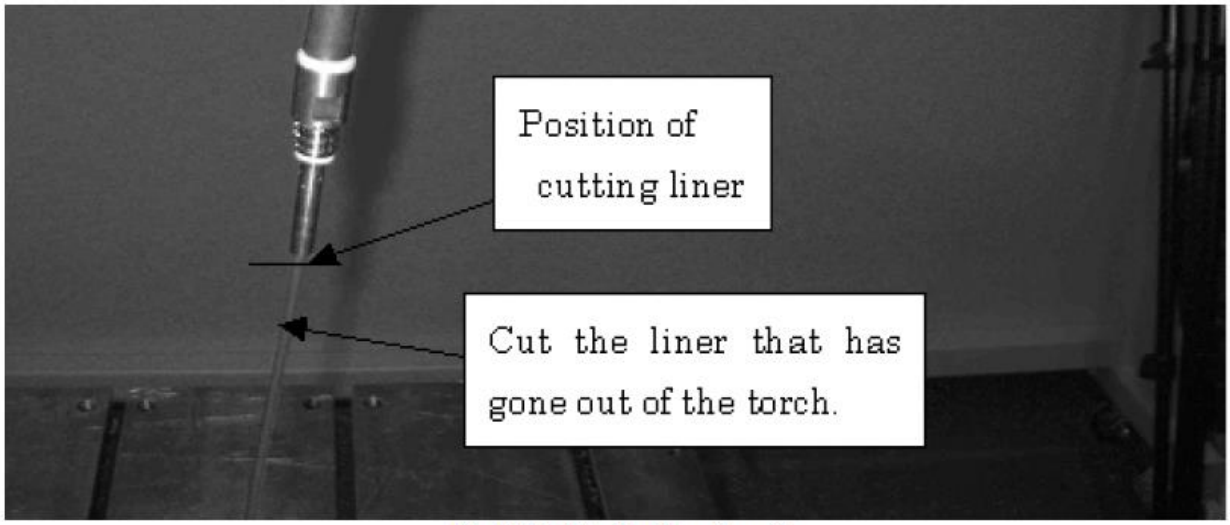


Fig.9.2.2 (b) Cutting liner 1

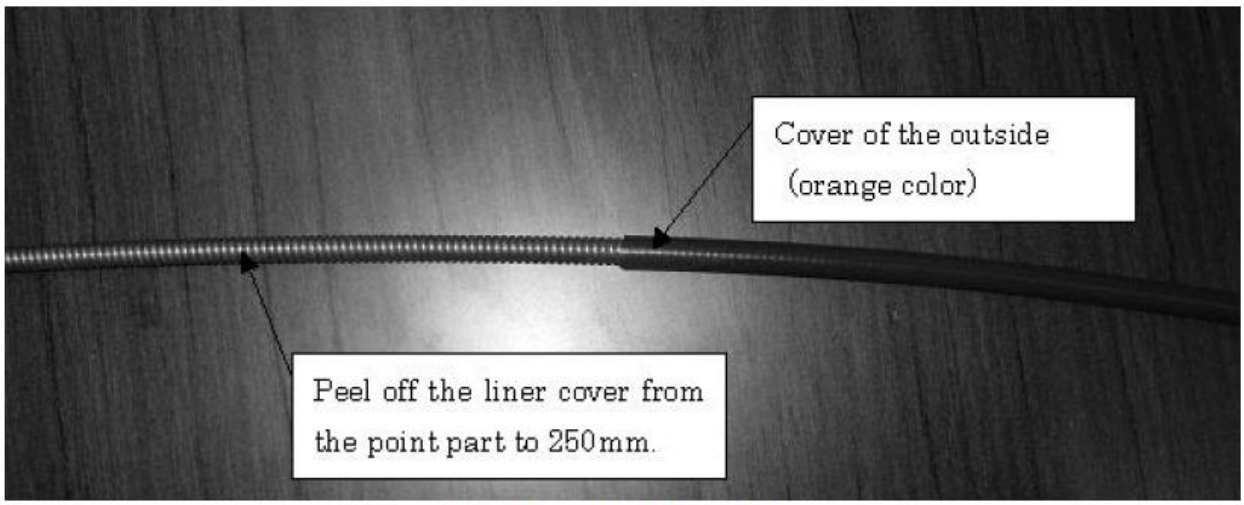


Fig.9.2.2 (c) Cutting liner 2

9.2.3 Replacing drive roll and wire guide

This chapter indicates procedure of replacing both of drive roll and wire guide(inside).

1 Loosen the screw in two places of the wire guide (outside) and remove wire guide (outside)
2 Knock down the idol arm below, and lift the drive roll of upper side.(See.Fig.9.2.3 (a))
3 Turn a triangular ring that is attached to the drive roll by the hand and phase it as unlocked.(See Fig.9.2.3 (b))
4 Remove the drive roll.
5 Remove wire guide (inside) after removing the 4 drive rolls.
6 Replace drive roll and wire guide (inside) to new one by opposite procedure. Pay attention to the direction of wire guide (inside) (See Fig.9.2.3 (c))
Turn at the same time and install two screws when you install the wire guide(outside).

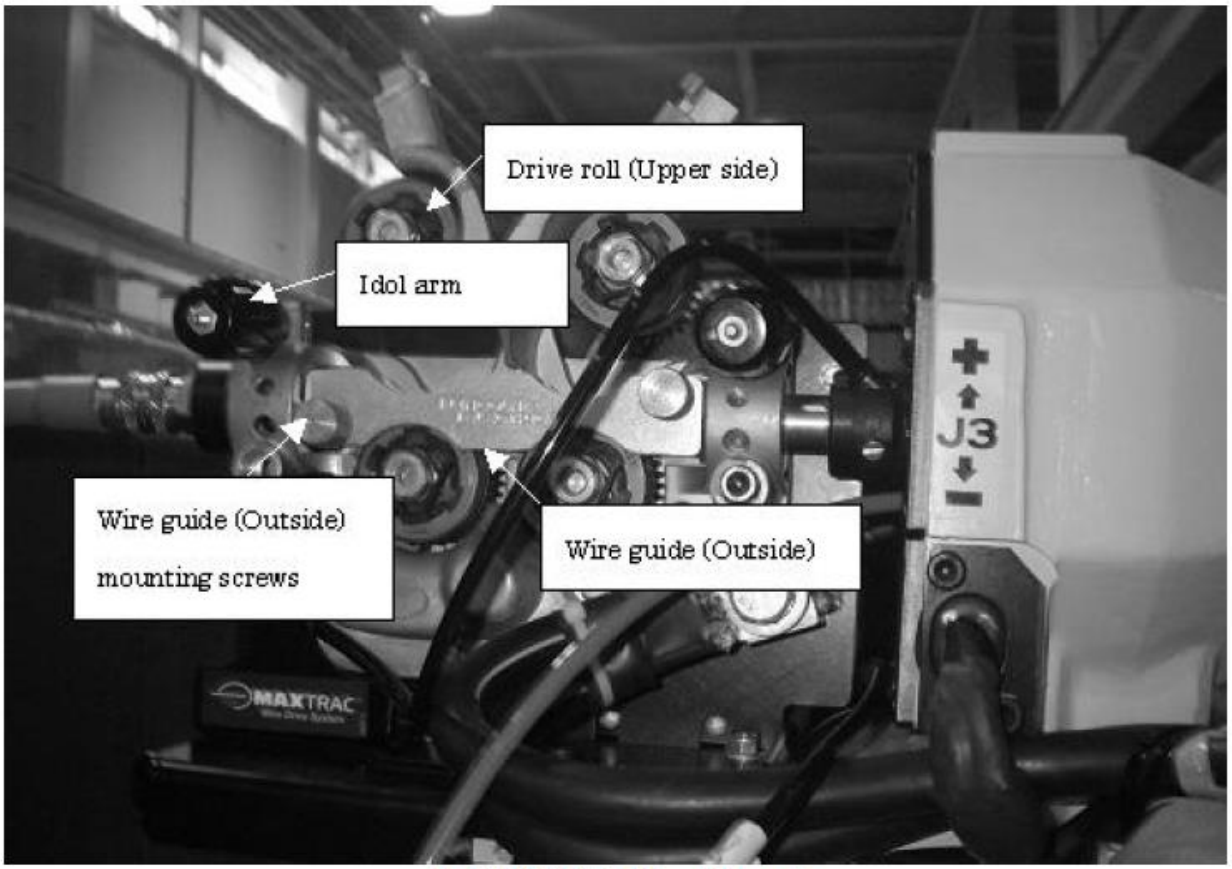


Fig.9.2.3 (a) Lift the drive roll

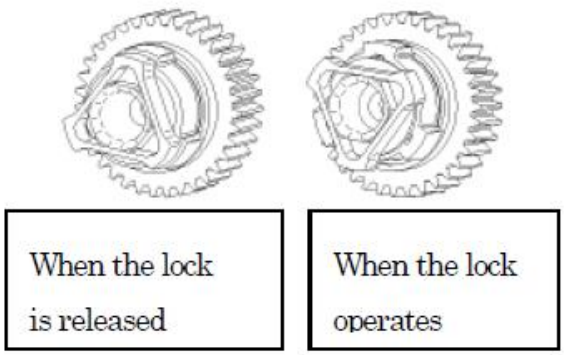


Fig.9.2.3 (b) Release lock of the drive roll

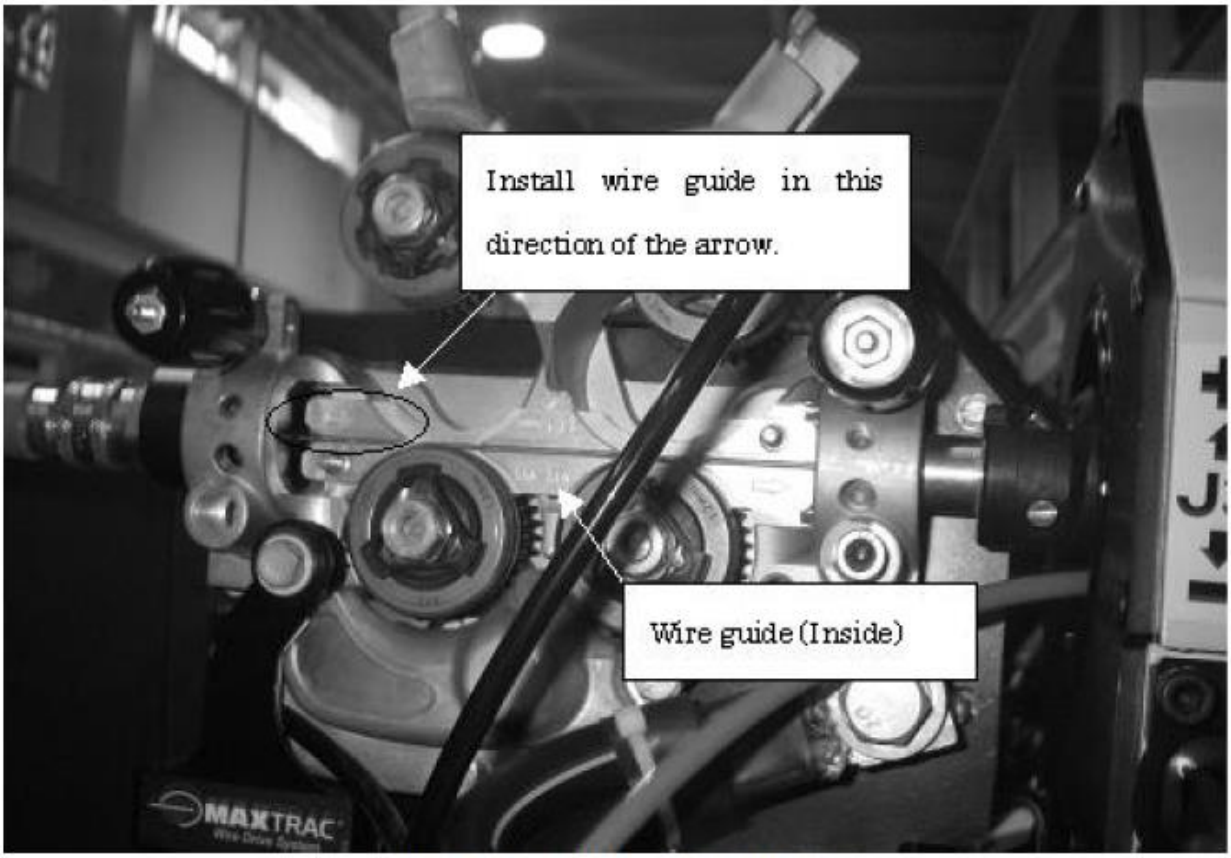


Fig.9.2.3 (c) Wire guide(Inside)

9.2.4 Replacing torch neck

When torch neck is broken ,replacing is needed. In this case, you may need to set TCP again by using, for example, torch recovery.
Use the procedure below for replacement.

1 Remove the nozzle.
2 Loosen screw of insulation cylinder by the monkey spanner and remove it. (See Fig.9.2.4 (a))
3 Loosen the screw of tip holder and remove tip holder. (See.9.2.4 (b))
4 Loosen insulation cap by the hand and remove it. (See.Fig.9.2.4 (c))
5 Remove rubber cover . (See.9.2.4 (d))
6 Loosen counter nut by the monkey spanner , remove torch neck and replace it to new one. (See Fig.9.2.4 (d)) Tighten counter nut by regulated torque.
7 Attach the tip holder. (See Fig.9.2.4 (b))
8 Attach rubber cover . (See.9.2.4 (d))
9 Attach insulation cap.(See Fig.9.2.4 (c))
10 Attach insulation cylinder. (See Fig.9.2.4 (a))
11 Attach the nozzle.

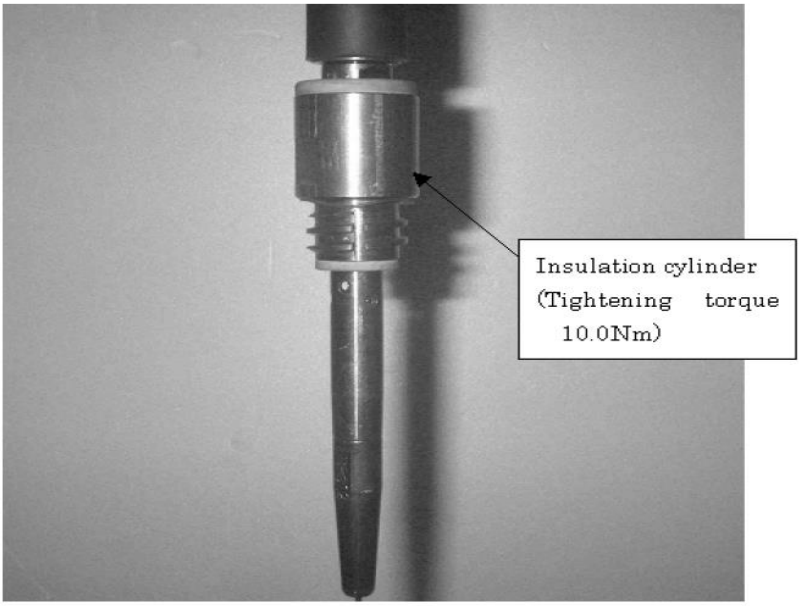


Fig.9.2.4 (a) Remove insulation cap

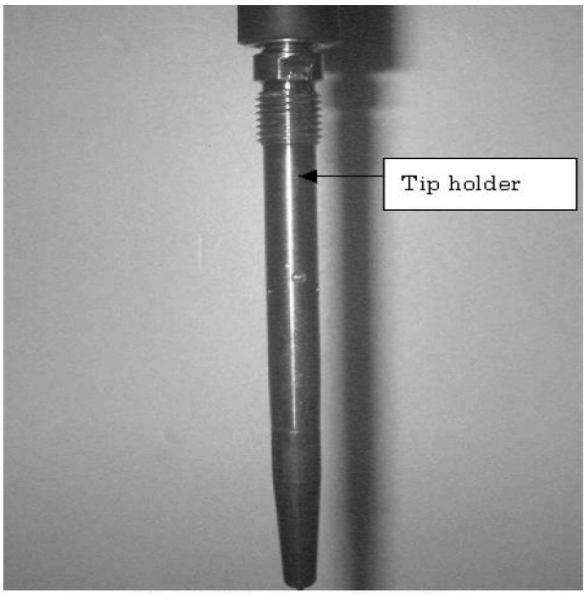


Fig.9.2.4 (b) Remove tip holder

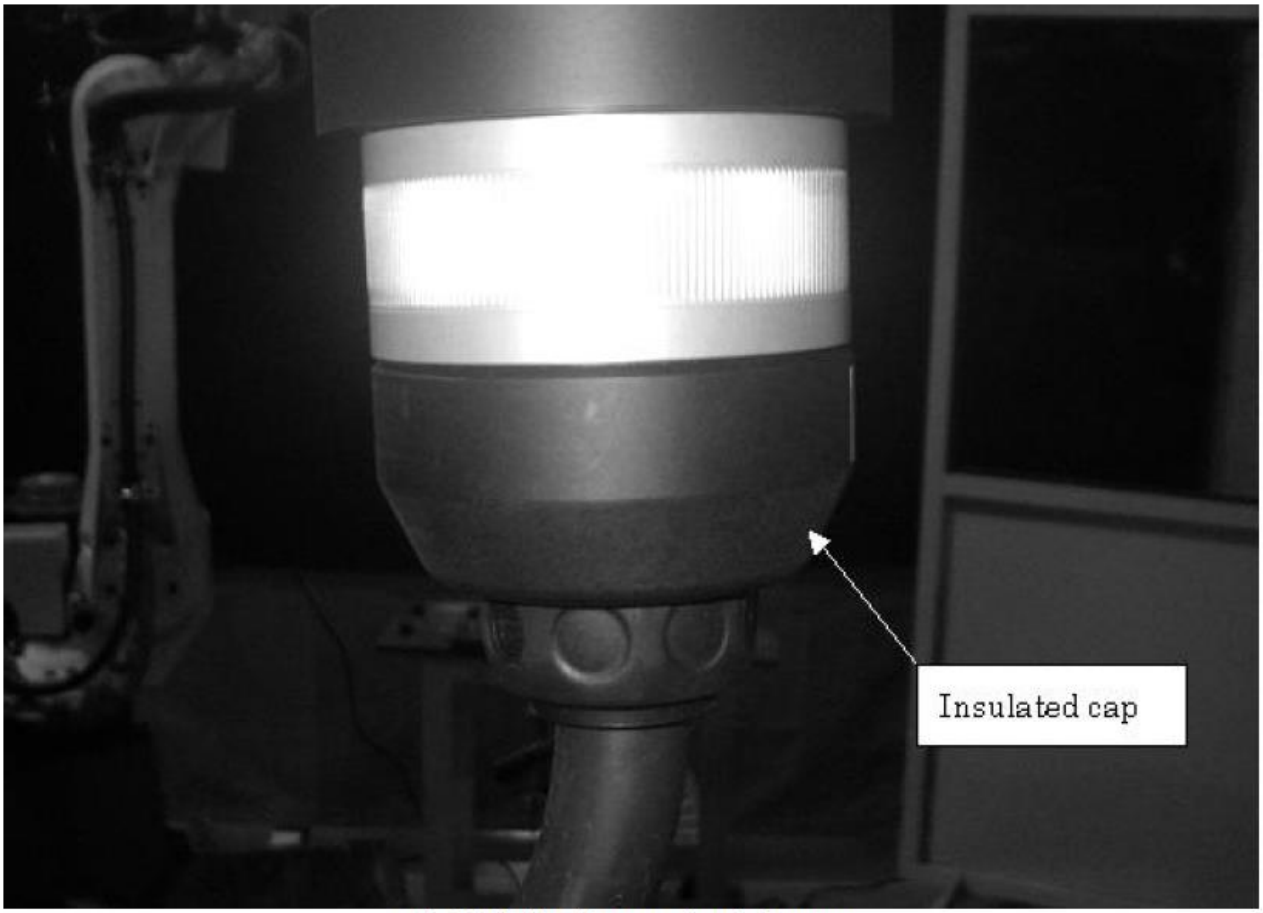


Fig.9.2.4 (c) Remove insulation cap

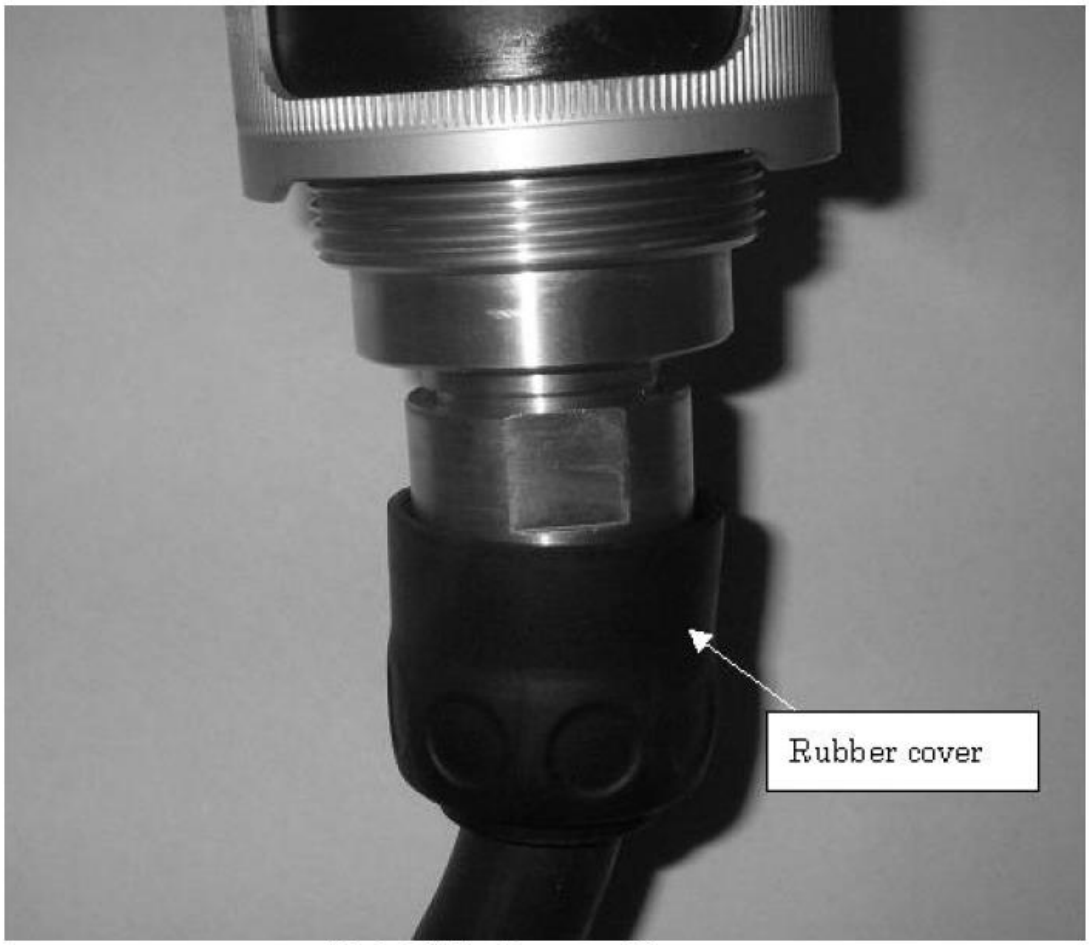


Fig.9.2.4 (d) Remove rubber cover

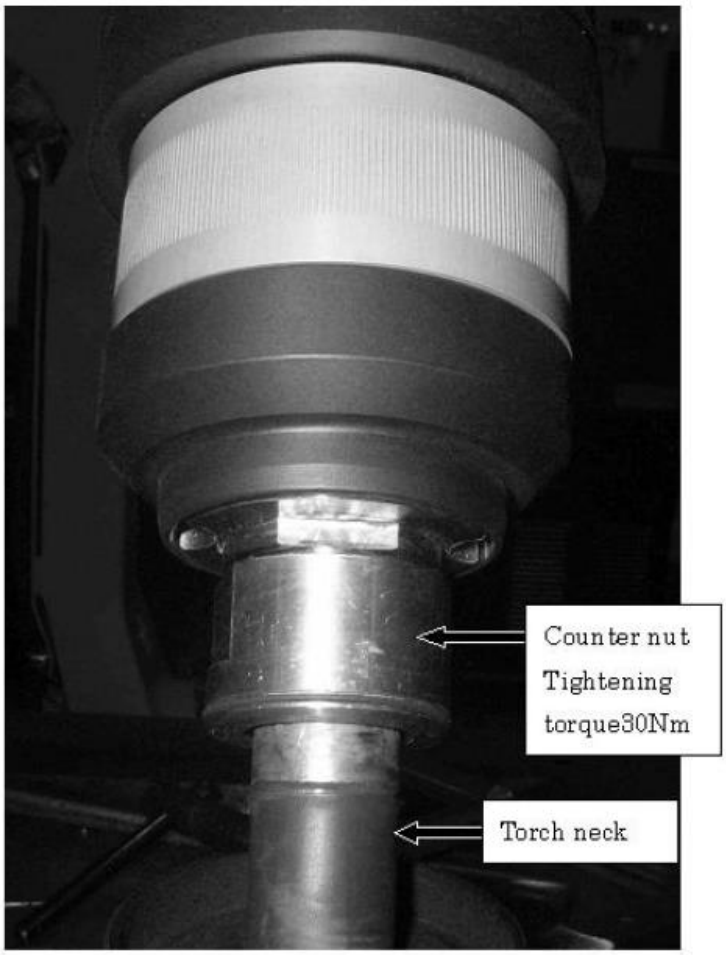


Fig.9.2.4 (e) Remove torch neck